

1. Which members of the Circle class are encapsulated?
Are the ones marked private.
2. What name must the constructor of a class have?
The name of the constructor has to be the same as the class name.
3. Explain the difference between the private and public access modifiers.
Private can only be accessed inside of the same class, public can be called or used by everyone, accessible from anywhere in the program.
4. Consider the following code. Is the last statement valid or invalid? Explain.
Circle dot = new Circle(2);
dot.radius = 5;
If the radius is private, then it's invalid. If it's public, it's valid.

5. Use the following class to answer the questions below:

```
public class Roo {  
    private int x;  
    public Roo {  
        x = 1;  
    }  
    public void setX(int z) {  
        x = z;  
    }  
    public int getX() {  
        return(x);  
    }  
    public int calculate() {  
        x = x * factor();  
        return(x);  
    }  
    private int factor() { return(0.12);  
    }  
}
```

- a) What is the name of the class? Roo
- b) What is the name of the data member? x
- c) List the accessor method. getX
- d) List the modifier method. setX
- e) List the helper method. private int factor()
- f) What is the name of the constructor? Roo
- g) How many method members are there? 5 (constructor, modifier, accessor, helper, normal)

6. What is the difference between a class and an object?

A class is the blueprint and an object is created from the class.

```
9. Public class Moo {  
private double y;  
private static int x;  
private static final z;  
...
```

- a) Which data member is a constant? **z**
- b) Which data members are variables? **y x**
- c) Which data member(s) are instance members? **y**
- d) Which data member(s) are class members? **x z**

11. Compare and contrast overriding methods to overloading methods.

Overriding: same method name, different parameters, same class.

Overloading: same method name, same parameters, different classes.